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PROFILE

Mechanical Engineering student at Bournemouth University building a torque-controlled, dynamically-walking bipedal robot for my MSc dissertation spanning mechanical design, embedded electronics, and robotics software (ROS2, control theory, reinforcement learning). Previously contributed to the design and validation of a hydrogen fuel cell system for a heavy-duty truck at Segula Technologies, working with Stellantis across a cross-functional team in the US, Germany, France, and Romania. I've since broadened that foundation with maintenance engineering internships across food processing and fertiliser manufacturing.

TECHNICAL SKILLS

- **SOFTWARE:** Catia v5/v6, SolidWorks, Fusion 360, Autodesk Inventor, ANSYS Mechanical, Abaqus
- **PROGRAMMING:** Python, C++.
- **CMMS:** Optimaint, Oracle.
- **ROBOTICS & CONTROL:** ROS2, Gazebo, Pinocchio, Isaac Lab, CAN bus, ODrive motor control, Embedded systems, Linux, Trajectory Planning, Inverse Kinematics.
- **ENGINEERING SKILLS:** Rapid Prototyping, 3D Printing, Tolerance stack Analysis, Mechanical drawings, Geometric dimensioning and tolerancing, Instrumentation Drawings, FMEA, Material Behaviour, Life cycle analysis, DFM.

Master of Science in Mechanical Engineering

September 2025 - September 2026

Bournemouth University, England

- Relevant coursework in Advanced Structural Mechanics, Failure Analysis and Prevention, Robotic Control Design and Model Based Engineering.

Master of Science in Mechanical Engineering: Maintenance and Quality

September 2021 - July 2023

Polytechnic School of Agadir, Morocco

- Thesis on "Improving the Availability of the PS III Sulphuric Acid Line through the Implementation of a Shell Replacement Procedure."

Bachelor of Science in Maritime Mechanical Engineering

September 2018 - July 2021

Higher Institute of Maritime Fisheries, Morocco

- Thesis on "Design and Implementation of a PID Temperature Control System for a Sardine Canning Cooking Installation."

WORK EXPERIENCE

Design Release Engineer, Segula Technologies

November 2023 - August 2024

- Contributed to the development of a hydrogen fuel cell system for a heavy-duty truck for Stellantis, collaborating with a cross functional team across the US, Germany, France, and Romania.
- Simplified system architecture by supporting the removal of the turbine bypass, reducing cost by €32.50 per part and lowering system complexity without compromising durability.
- Supported supplier issue resolution on components, such as fuse integration, temperature sensors, engine mounts, and subsystem mounting points.
- Enhanced vehicle safety by validating the "eBurn" function, an energy-dissipation strategy that prevents brake overheating and maintains regenerative braking on long downhill grades.
- Participated in testing and validation activities, analysing failures (e.g. Hydrogen inlet pressure valve) and helping define improvement measures.

Graduate Mechanical Engineering Internship, OCP Group

February 2023 - May 2023

- Led a root cause analysis study of critical production line failures, applying fishbone diagrams and 5 why's to identify and eliminate recurring issues.
- Created a 3D model of the Heat Recovery System Tower in SolidWorks and ran a static structural simulation in ANSYS Mechanical to evaluate external constraints.
- Developed and implemented a streamlined shell replacement procedure for the HRS Tower, reducing downtime and improving operational efficiency.
- Delivered a 15% increase in overall availability of the PS III sulphuric acid line by integrating simulation insights with optimised maintenance strategies.

Design & Manufacturing Engineering Internship, ENSA Mechanical Laboratory

June 2022 - August 2022

- Designed a 3D model of an educational torsion test bench in SolidWorks, applying reverse engineering principles such as systematic decomposition and functional analysis to replicate and enhance the original design.
- Upgraded the measurement system by replacing the external comparator with a graduated pulley, improving accuracy and ease of use.
- Operated both conventional and CNC machining equipment to manufacture and assemble project components.
- Achieved an 80% reduction in setup time for the test bench through the redesigned measuring system.

PROJECTS

Bipedal Robot MSc Dissertation

March 2026 – Present

Funded by Bournemouth University

Designing and building a 5-link, four-actuator planar bipedal robot for my MSc dissertation, extending a published single-leg design with custom mechanical, electrical, and software architecture.

- Built full ROS2-based robot architecture from scratch, including URDF kinematic modeling, trajectory planning, and inverse kinematics for the leg.
- Integrated brushless motor actuation (ODrive controllers, CAN bus) with closed-loop position control, validated through iterative prototyping, 3D printing, and bench testing.
- Progressing toward dynamic walking via torque-based control, comparing classical control (Hybrid Zero Dynamics) against reinforcement learning (Isaac Lab) under shared performance metrics.

Life Cycle Management Project

September 2025 – January 2026

Improving the carbon footprint of a Roll-A-Car 740 Mechanical Jack

Life Cycle Management project focused on improving the environmental impact of a mechanical lifting jack through quantified life cycle analysis and redesign on SolidWorks.

- Established a baseline carbon footprint of around 560 kg CO₂, primarily driven by material choice and manufacturing processes.
- Ran a topology optimisation on ANSYS while preserving 55% to 60% of the starting mass.
- Re ran a static structural analysis to validate that the performance was maintained post redesign.
- Reduced total carbon footprint to 440 kg CO₂, achieving an approximate 22% reduction.

3D-Printed Robotic Arm (3DOF → 5DOF → 6DOF)

October 2024 – August 2025

- Designed and iteratively developed a robotic arm from 3DOF to a 6DOF system, focusing on precision, torque, and efficiency through evolving drive technologies (planetary gearboxes → hybrid cycloidal-belt transmissions).
- Optimised manufacturability and assembly, reducing hardware requirements by using standard ball bearings (608-2z), stepper motors (Nema 17), and bolts. Ensured parts were low cost, easy to print, and reliable on consumer-grade 3D printers.
- Implemented a two-layer control system: a high-level layer (ROS + MoveIt in a VM) generated joint angles from user-defined RViz poses via inverse kinematics, while a low-level Python layer translated those angles into real-time stepper motor commands.

ADDITIONAL INFORMATION

- **Languages:** English (Professional working proficiency), French (fluent), Arabic (native).
- **International experience:** Academic and professional experience across Morocco, the UK, and multinational engineering teams in Europe and North America.
- **Driving licence:** International Driving Permit valid until September 2026 and currently working on getting a full UK licence.